Yixuan Shi

Max Planck Institute for Tax Law and Public Finance
Marstallplatz 1, 80539 Munich, Germany
Yixuan.Shi@tax.mpg.de

Research Fields

Applied Microeconomic Theory; Behavioural and Experimental Economics; Public Economics.

Employment

Senior Research Fellow (Postdoctoral) 2026–present
Max Planck Institute for Tax Law and Public Finance, Munich
Department of Public Economics

Education

Ph.D. in Economics 2020–2026
Ludwig-Maximilians-Universität München (LMU Munich)
Max Planck Institute for Tax Law and Public Finance

MPhil in Economic Research, University of Cambridge 2018–2019

MSc in Economics, University College London (Distinction) 2017–2018

BSc(Econ) in Economics, University College London (First Class Honours) 2014–2017

Publications

Dynamic Volunteer's Dilemma With Procrastinators [Published version]
Journal of Public Economic Theory, 2025, 27(2), e70027.

Abstract. Only one trip planner is needed for a group of friends to enjoy a pleasant trip and only one country is needed to coordinate on international talks that can be beneficial for all participating countries. We study a dynamic volunteering dilemma game in which two players choose to volunteer or wait given there have not been any volunteering actions in the past. The players can be procrastinators and the benefits of volunteering arrive later than the costs. We fully characterize the stationary Markov Strotz-Pollak equilibria. When the cost of volunteering is sufficiently small or agents' present-bias parameters are sufficiently close, there always exists an equilibrium in which both players randomize. This equilibrium features stochastic delay, and the delay is exacerbated if one or both agents become more present-biased. However, if the agents differ significantly in their present-bias parameters, this difference may act as a 'natural coordination device' and the unique stationary equilibrium predicts that only the less severe procrastinator volunteers, and this may result in an even quicker provision compared with the case of two exponential discounters.

Working Papers

Two Routes to Office: Campaign Contests in Mixed Electoral Systems [Paper]

Abstract. In mixed electoral systems with dual candidacy, candidates can win parliamentary seats either in district elections or through party lists. Limited campaign budgets create a trade-off between individual and team competition: personalised campaigning strengthens candidates' personal appeal and district prospects, while party-centric campaigning increases their party's electoral support and overall seat share. We develop a contest-theoretical model in which candidates allocate resources between the two contests. Under a closed list, predetermined rankings generate a division of labour in the equilibrium: lower-ranked candidates devote their resources to district campaigning, while higher-ranked candidates divide resources between two contests. Individual competition further shapes these allocations: they depend on the rankings of candidates' district opponents and, in larger parties, on the opponents faced by their teammates.

Sign Stealing in Team Contests: Coordination, Unpredictability, and Competition Intensity [Paper]
with Stefano Barbieri and Sven A. Simon

Abstract. Communication can facilitate coordination within teams, but it may also leak and reveal intentions to the opponent, as when a baseball team's signs are stolen. We study how this tension shapes behaviour in a team contest, theoretically and in a laboratory experiment. Our theory predicts that secure communication plays a dual

role: it improves internal coordination while concealing intentions from the opponent, thereby raising competition intensity, measured by effort, for all parties. The experiment provides causal evidence for these effects. When communication leaks, its concealment role disappears and teammates trade off coordination against unpredictability. We focus on the case in which coordination concerns dominate, so that in theory teammates should still communicate; yet they sometimes choose not to, producing costly coordination failures. Even so, effort remains close to its level under secure communication and well above its level without communication. Theory and experiment agree that jamming an opponent's communication may be more effective than intercepting it.

Play Nice, Party Hard [Paper]
with Afiq bin Oslan
Revise and resubmit, *Journal of Theoretical Politics*
Abstract. Political parties often appear united despite containing factions with divergent preferences. We study why party defections are rare by opening the intra-party 'black box' through a set of game-theoretical models. In the model, the leadership sets the party's policy platform anticipating exit threats from co-partisan rival factions. Rivals weigh policy proximity against the benefits of belonging to a larger party: smaller factions lack credible exit threat and remain loyal, while larger factions may require policy compromises. We emphasise the role of party size and show that unity can be sustained through a snowballing mechanism: once the leadership compromises and secures the loyalty of some factions, other factions become more willing to remain—even with no political concessions—because doing so places them with a larger and more powerful party. Leaders need only win part of the party to induce broader unity, implying that even internally diverse parties can remain stable.

Give One Way or Another? Choosing Between Public Goods and Private Transfers Under Inequality [Paper]
with Anwesha Banerjee and Natalie Struwe
Abstract. We study how public good contributions and private transfers interact as channels of giving in unequal groups. Building on a model of altruistic preferences, we conduct a pre-registered, between-subjects experiment (n = 478) that varies income inequality and the availability of simultaneous private transfers in a repeated, two-person, non-linear public good game. Allowing transfers significantly reduces public good contributions across all tested inequality levels, as participants substitute between the two channels. Transfers occur even among equally endowed participants and from poorer to richer participants, with exploratory analysis suggesting a role for positive reciprocity. In highly unequal groups, allowing transfers improves poorer participants earnings without reducing those of richer participants. These findings highlight how private transfers can reduce within-group inequality without sacrificing aggregate earnings, despite crowding out public good contributions.

Work in Progress

Me or We? Incentive Design and Specialisation in Overlapping Contests
Abstract. When teammates also compete for individual recognition, how should a leader divide a team prize? We study a team contest in which team leaders commit to fixed team-prize division before members split their effort between a team contest and an individual contest among all members. Even with symmetric agents, every pure-strategy equilibrium concentrates each team's entire prize to one member. Once a member with a small share focuses entirely on the individual battle, taking the rest of her prize share costs no team effort and strengthens the remaining contributor. Evenly matched teams give the whole prize to their stars. An outmatched team, however, may do the reverse, rewarding its non-star or diluting its contributor's prize share to steer efforts towards individual competition. The results identify individual competition as a source of endogenous specialisation within teams and show how competitive asymmetry changes both whom to reward and how concentrated the reward is.

Experience and Paternalistic Intervention
with Sven A. Simon and Leyi Tu
Abstract. We design an experiment to study how personal experience affects whether and how individuals intervene in others risky choices. Participants choose between lotteries with equal expected payoffs but different risks, with probabilities and payoffs fully known. We vary whether feedback on realised and forgone outcomes is received before or after the intervention decision. Participants can recommend an option, restrict another participants choices, or refrain from intervening, allowing us to examine the likelihood, direction, and intensity of intervention. We also explore possible channels, including emotional responses such as regret and rejoicing.

Sequential Volunteering under Deterioration
with Sacha Bourgeois-Gironde and Germain Lefebvre
Abstract. We study sequential public good provision when completion requires multiple costly volunteering actions and benefits deteriorate over time. In a symmetric interior equilibrium, individual volunteering rates are indepen-

dent of the provision stage at a given time, but aggregate volunteering slows as contributors exit. Each contribution extends the deadline for further participation, leaving early movers with the tightest viability constraint. For a fixed provision requirement, larger volunteer pools slow aggregate volunteering and tighten early-stage participation constraints, reducing the probability of completion.

Teaching Experience

International Public Economics 2021, 2023
Teaching Assistant, graduate course; LMU Munich
Winter semesters. Student evaluation: 1.2 (scale: 1 = best, 5 = worst).

Research Visits

Nuffield College, University of Oxford May 2025
Host: Zirui (Tony) Yang

Queensland University of Technology December 2024
Host: Changxia Ke

Conferences and Presentations

2026 Maastricht Behavioural and Experimental Economics Symposium; 15th Lisbon Meetings in Game Theory and Applications.

2025 18th RGS Doctoral Conference in Economics, Dortmund; 28th IPSA World Congress of Political Science, Seoul.

2024 15th Australasian Public Choice Conference, Brisbane; EDGE Jamboree, Dublin; 120th American Political Science Association Annual Meeting, Philadelphia; The Future of Contest Theory, Berlin.

2023 3rd Durham Economic Theory Conference, Durham; Micro Brown Bag Seminar, LMU Munich; 16th RGS Doctoral Conference in Economics, Bochum; Öffentliche Finanzen und Politische Ökonomie IX, Berlin.

2022 19th International Conference on Social Dilemmas, Copenhagen.

Academic Service

Organising Committee, "Strange Political Bedfellows" Workshop in Berlin 2026
with Afiq bin Oslan

Discussant, 2nd Berlin PhD Conference in Economics 2026

PhD Representative, MPI for Tax Law and Public Finance 2025

Electoral Committee, MPI for Tax Law and Public Finance 2020–2025

Referee, Journal of Public Economics

Summer Schools and Other Academic Activities

2025 8th Lindau Nobel Meeting in Economic Sciences.

2023 Max Planck Summer School on the Political Economy of Conflict and Redistribution, Berlin; Behavioural Game Theory Summer School, University of East Anglia, Norwich.

Research Tools and Languages

Research tools: Stata, R, oTree, Mathematica, MATLAB, Python, LaTeX.
Languages: English (fluent), Mandarin (native), German (B1).

References

Kai A. Konrad
Professor
Max Planck Institute for Tax Law and Public Finance
Kai-Konrad-Office@tax.mpg.de

Matthias Lang
Professor
Ludwig-Maximilians-Universität München
matthias.lang@econ.lmu.de

Rabah Amir
Professor
University of Iowa
rabah-amir@uiowa.edu